

# Jeewant Choudhry

M.S. Computer Engineering • Hardware Engineer

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## Education

Virginia Polytechnic Institute and State University (Virginia Tech)

M.S. IN COMPUTER ENGINEERING • GPA: 3.95/4.00

Blacksburg, Virginia

Aug. 2025 - May 2027

• **Relevant Courses:** Cyber Physical Systems, Quantum Engineering, Network Security, Advanced Machine Learning, Applications of Machine Learning

Manipal Institute of Technology, Manipal

B.TECH. IN ELECTRONICS AND INSTRUMENTATION ENGINEERING

Kamataka, India

2021 - 2025

## Work Experience

VIRGINIA TECH

RESEARCH ASSISTANT AND ECE CAMPUS GRADUATE AMBASSADOR (SUMMER '26)

Blacksburg, Virginia

May 2026 - Present

- Designed a **UWB-based proximity alert system** for paving crews, using two-antenna PDoA for 10 cm ranging and  $\pm 5^\circ$  bearing, eliminating a planned 60 GHz mmWave radar and cutting per-unit BOM and power draw.
- Building a **ranging-capacity model** showing one anchor sustains  $\sim 30$  cones within a sub-5s alarm budget, and specified LiFePO4 power, IP67 enclosures, and FCC modular-certified parts for field deployment.
- Supporting **600+ students** through office hours, email communication, and project follow-ups, helping keep student questions and course updates organized.

TEACHING ASSISTANT (SPRING '26) | GRADER (FALL '25)

Sep. 2025 - Apr. 2026

- Coordinated instructional support across **3 graduate courses serving 160+ students**, holding weekly office hours, resolving queries, and developing a **most useful concepts checklist** for analytics, finance, and engineering topics.
- Facilitated assignment tracking and faculty communication to keep course operations organized and aligned with course goals.

GOLDMAN SACHS

DATA ANALYST, ASSET AND WEALTH MANAGEMENT

Bengaluru, India

Jan. 2025 - Aug. 2025

- Executed **User Acceptance Tests (UATs)** across Private Wealth Management systems using SQL and Python, structured test cases, business datasets, and defect logs to identify **10+ issues** before production and support release readiness for critical workflows.
- Built a **Tableau-based business intelligence dashboard** for senior stakeholders in the Client Solutions Group, tracking **missed mandates** across asset class, client segment, and regional views to improve visibility into lost business opportunities.
- Automated **Jira project reporting** through an Alteryx workflow that tracked ticket updates, project completion, and overdue tasks, improving documentation completion rates by **30%** across more than **15 projects**.

INDIAN INSTITUTE OF TECHNOLOGY, BHU

RESEARCH INTERN, ELECTRONICS DEPARTMENT

Varanasi, India

May 2024 - Jul. 2024

- Designed and validated **4+ stepper** and **DC motor driver circuits** with customized control logic in the **Circuits and DC Lab**, supporting **high precision motion control** and improving the reliability of motor control setups used for hardware testing.
- Conducted extensive validation of **10+ IC chips** and optimized **5+ PCB layouts** using MATLAB and industry-standard design tools, achieving **>90% testing accuracy** and improving **signal quality by 30%** to make circuit behavior more stable during repeated testing.
- Evaluated **speed, direction, and incremental position control** across both motor setups through iterative hardware testing, improving **system integration, prototype stability**, and overall **motor control** in precision electromechanical setups.

## Key Projects

Precision Motion Control Prototype

ALTIUM DESIGNER • MATLAB • ARDUINO MEGA • 555 TIMER • MOSFETS • OPTOCOUPERS

- Built a precision motor control prototype for **DC** and **Stepper actuation** by integrating **555 timers**, **MOSFETs**, and **optocouplers** into customized driver stages, with the goal of improving **switching reliability** and **hardware robustness** in lab-scale applications.
- Bench tested the customized driver stages across both motors, **refining component interfacing** and control behavior through iterative hardware testing to improve stability and overall **motion control performance**.

Real-Time Energy Monitoring System

ARDUINO • ESP8266 • VOLTAGE/CURRENT SENSING • EMBEDDED C • WI-FI • DASHBOARD VISUALIZATION

- Prototyped and tested an **IoT-enabled smart energy meter** using **Arduino** and **ESP8266 Wi-Fi**, enabling real-time monitoring of voltage, current, and power usage through an embedded sensing and communication setup for continuous electrical data tracking.
- Constructed a connected monitoring workflow that sent live electrical readings to a dashboard visualization layer, improving system visibility and allowing **remote monitoring of power metrics** across the connected system environment.

Electric Vehicle Feasibility Model

EXCEL • PYTHON • PANDAS • NUMPY • MATPLOTLIB • KPI REPORTING • UNIT ECONOMICS • PRICING ANALYSIS

- Built and validated a **market-sizing and unit-economics model** for an electric vehicle venture on campus using Python, Excel, pricing analysis, and KPI dashboards to drive around **8-12% margin lift** and nearly **20% faster turnaround**.

## Skills

Languages

Electrical & Systems

Data & Reporting

Tools & Workflow

Python, SQL, C++, MATLAB, Embedded C

Circuit Design, PCB Layout, Motor Drivers, Motion Control, Arduino, ESP8266, Sensors and Actuators

Alteryx, Tableau, Dashboard Development, Data Validation, User Acceptance Testing

Altium Designer, Jira, Confluence, Slack, Git